

the speed of your Internet and the traffic on the Twitter server, you will get an R list class object responses within a few minutes and an R list class object.

```
>tweetList<- searchTwitter("#love",n=500)
>mode(tweetList)
[1] "list"
>length(tweetList)
[1] 500
```

In R, an object *list* is a compound data structure and contains all types of R objects, including itself. For further analysis, it is necessary to investigate its structure. Since it is an object of 500 list items, the structure of the first item is sufficient to understand the schema of the set of records.

```
>str(head(tweetList,1))
List of 1
 $ :Reference class 'status' [package "twitterR"] with 20 fields
   ..$ text : chr "https://t.co/L8dGustBQX #SaveOne #LLOVE
   #GotItWrong #JCole #Drake #Love #F4F #follow #follow4follow
   #Repost #followback"
   ..$ favorited : logi FALSE
   ..$ favoriteCount : num 0
   ..$ replyToSN : chr(0)
   ..$ created : POSIXct[1:1], format: "2017-10-04
   06:11:03"
   ..$ truncated : logi FALSE
   ..$ replyToSID : chr(0)
   ..$ id : chr "915459228004892672"
   ..$ replyTUID : chr(0)
   ..$ statusSource : chr "<a href='\"http://twitter.com\"'
   rel='\"nofollow\"'>Twitter Web Client</a>"
   ..$ screenName : chr "Lezzardman"
   ..$ retweetCount : num 0
   ..$ isRetweet : logi FALSE
   ..$ retweeted : logi FALSE
   ..$ longitude : chr(0)
   ..$ latitude : chr(0)
   ..$ location :chr "Bay Area, CA, #CLGWORLDWIDE <ed><u>#0A0B
   <u>#0B0D<ed><u>#0B2<u>#0A0F">
   ..$ language : chr "en"
   ..$ profileImageUrl:chrhttp://pbs.twimg.com/profile_
   images/444325116407603200/xm292Dv6_normal.jpeg"
   ..$ urls : 'data.frame': 1 obs. of 5
   variables:
    ..$ url : chr "https://t.co/L8dGustBQX"
    ..$ expanded_url: chr "http://cdbaby.com/cd/saveone"
    ..$ display_url : chr "cdbaby.com/cd/saveone"
    ..$ start_index : num 0
    ..$ stop_index : num 23
   ..and 59 methods, of which 45 are possibly relevant:
    .. getCreated, getFavoriteCount, getFavorited, getId,
    getIsRetweet,
```

```
.. getLanguage, getLatitude, getLocation, getLongitude,
getProfileImageUrl,
.. getReplyToSID, getReplyToSN, getReplyTUID,
getRetweetCount,
.. getRetweeted, getRetweeters, getRetweets,
getScreenName, getStatusSource,
.. getText, getTruncated, getUrls, initialize, setCreated,
setFavoriteCount,
.. setFavorited, setId, setIsRetweet, setLanguage,
setLatitude, setLocation,
.. setLongitude, setProfileImageUrl, setReplyToSID,
setReplyToSN,
.. setReplyTUID, setRetweetCount, setRetweeted,
setScreenName,
.. setStatusSource, setText, setTruncated, setUrls,
toDataFrame,
.. toDataFrame#twitterObj
>
```

The structure shows that there are 20 fields of each list item, and the fields contain information and data related to the tweets.

Since the data frame is the most efficient structure for processing records, it is now necessary to convert each list item to the data frame and bind these row-by-row into a single frame. This can be done in an elegant way using the *do.call()* function call, as shown here:

```
loveDF<- do.call("rbind",lapply(tweetList, as.data.frame))
```

Function *lapply()* will first convert each list to a data frame, then *do.call()* will bind these, one by one. Now we have a set of records with 19 fields (one less than the list!) in a regular format ready for analysis. Here, we shall mainly consider 'created' field to study the distribution pattern of arrival of tweets.

```
>length(head(loveDF,1))
[1] 19
>str(head(loveDF,1))
'data.frame': 1 obs. of 19 variables:
 $ text : chr "https://t.co/L8dGustBQX
 #SaveOne #LLOVE #GotItWrong #JCole #Drake #Love #F4F #follow
 #follow4follow #Repost #followback"
 $ favorited : logi FALSE
 $ favoriteCount : num 0
 $ replyToSN : chr NA
 $ created : POSIXct, format: "2017-10-04
 06:11:03"
 $ truncated : logi FALSE
 $ replyToSID : chr NA
 $ id : chr "915459228004892672"
 $ replyTUID : chr NA
 $ statusSource : chr "<a href='\"http://twitter.com\"'
 rel='\"nofollow\"'>Twitter Web Client</a>"
```